

Ever since Thomas Edison's phonograph belched out the words "Mary had a little lamb", the world has stayed in a relentless pursuit of recording and reproducing audio. The advancements led to more intricate machines, specializing in different aspects of sound. How do all these machines interact to produce our favourite symphonies? Cables of course. And even though we're moving towards a wireless society, the premium fidelity provided by wires still stands leagues ahead.

Cable connections as we know are broadly split in two categories; balanced and unbalanced. Unbalanced connections have two conductors - one for carrying the signal and the other for grounding. Balanced on the other hand has three conductors - one stays for grounding and the other two carry the signal. One of those two carries the signal in normal polarity while the other carries it with the polarity flipped and at the destination is corrected back to normal. When the second signal is inverted to original orientation, all of the noise it captured also gets its polarity flipped. Now when the two signals meet, each carrying noise of opposing polarities, the noise gets nullified, resulting in clean audio. This makes balanced cable an ideal option for long cable networks.

The cable connections consist of a wire and connector ends, with each of them having their own abilities and feature sets.

First up, the most common audio connector known to mankind - Phone Connector. Never heard of it? Well you sure have used it, in fact you use it every day. The quarter inch TRS connector or the commonly designated 'headphone cable', both belong to the parentage of phone connectors. For a bit of history, when telephones were first brought to mainstream usage, they came with a middleman called the 'operator'. The operator's job was to manually connect you to the other person with a phone connector. Fast forward to today, we have them in three different sizes and for three different cable types.

6.35, 3.5 and 2.5

6.35 mm connectors are most notably associated with electric guitars as the instruments come with a female side of the connector. They are also found on certain loudspeakers as an alternative to Speakon connectors (discussed later) and professional headphones.

3.5 mm connectors are the ones used in almost every mobile device. They are the ones which come on the end of most headphones, earphones